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Open Computational Chemistry (OCC) - A portable software library and program for quantum chemistry and crystallography

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Software

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Summary

Open Computational Chemistry (OCC) is a modern software library designed for calculating the properties and electronic structure and interactions of molecules and molecular crystals. The software serves a dual purpose: it provides an open, accessible platform for researchers developing new computational methods, while also being production-ready and fast enough for relatively large-scale calculations on personal computers and high-performance computing systems alike.

Many traditional computational chemistry programs require complex installation procedures or rely on large dependencies can only run on specific operating systems. OCC is designed as a priority to be portable: it runs on Windows, macOS, and Linux, provides interfaces for C++, Python, and JavaScript, and with the capability to even run entirely within web browsers through WebAssembly (WASM), making computational chemistry accessible without specialized hardware or software installation. OCC is already widely used as the primary computational backend of CrystalExplorer(Spackman et al., 2021), a widely-used graphical program for crystal structure analysis, providing capabilities such as automatic calculation of interaction energies between molecules in crystals, accurate determination of total crystal energies with automatic convergence checking, and evaluation of electron density distributions.

Statement of need

The landscape of computational chemistry software is dominated by established packages that, while powerful, often suffer from legacy codebases, limited portability, and barriers to modification. Most existing quantum chemistry software requires complex installation procedures and system-level dependencies, limiting accessibility for educational purposes and rapid prototyping. Furthermore, the integration between quantum chemistry and crystallography remains fragmented across different specialized tools, requiring researchers to chain together multiple incompatible programs for multi-scale workflows.

OCC addresses these challenges as an open, portable, and modifiable platform for computational chemistry and crystallography. The software serves dual purposes: as a computational backend for production applications like CrystalExplorer, a widely-adopted graphical tool for crystal structure analysis, and as a research platform for developing new computational methods. This distinguishes OCC from purely academic codes (often limited in scope, modifiability or performance) and purely commercial alternatives (which lack transparency and modifiability).

Key features enabling this versatility include:

Portability and Accessibility: OCC is, to our knowledge, the first full-featured quantum

chemistry package that can run entirely in web browsers via WASM, enabling interactive calculations without installation or specialized hardware, on computers, tablets, phones or any device with a browser supporting WASM. Native builds support Windows, macOS, and Linux with comprehensive continuous integration testing, while language bindings for C++, Python, and JavaScript facilitate integration into diverse workflows.

Unified Framework: By integrating electronic structure methods with crystallographic analysis in a single modern C++ codebase, OCC eliminates the friction of multi-scale workflows. Unique capabilities include crystal growth free energy predictions bridging molecular-level calculations and macroscopic properties, automatic lattice energy convergence, and pairwise interaction energy calculations used in production by CrystalExplorer.

Open and Modifiable: The codebase follows contemporary software development best practices including comprehensive continuous integration testing across all platforms, automated releases for Python packages via PyPI, and Javascript via NPM and well-documented APIs with dependency management via CPM.cmake. This enables researchers to rapidly develop and test new methods. Unlike monolithic legacy codes, OCC's modular architecture facilitates extension and modification for specific research needs.

Implementation and Features

OCC implements Hartree-Fock and Density Functional Theory (DFT) with support for LDA, GGA, and meta-GGA functionals via libxc (Lehtola et al., 2018), density fitting (RI-JK) methods for coulomb and exchange interactions, implicit solvation via COSMO(Klamt & Schüürmann, 1993) and SMD(Marenich et al., 2009), and dispersion corrections (XDM(Becke & Johnson, 2007; Johnson & Becke, 2006; Otero-de-la-Roza & Johnson, 2012), DFTD4(Caldeweyher et al., 2017, 2019, 2020)). For crystallography, OCC provides CIF file processing via gemmi (Wojdyr, 2022), fast periodic bond detection, symmetry-unique molecule generation, CrystalExplorer model energies (Mackenzie et al., 2017; Spackman, Spackman, et al., 2023), and automatic pair-based lattice energy summation for neutral molecular crystals with symmetry.

Unique capabilities include crystal growth free energy predictions(Spackman, Walisinghe, et al., 2023) combining lattice energies, interaction energy decomposition, and vibrational/configurational entropy contributions. The distributed multipole analysis (DMA)(Stone, 2005) implementation provides multipole expansions up to hexadecapole level with GDMA-compatible output. WebAssembly compilation via Emscripten (Zakai, 2011) enables full-featured quantum chemistry calculations directly in web browsers, supporting interactive educational tools and client-side computational workflows without server infrastructure.

OCC employs a modular C++ architecture using Eigen3 (Guennebaud et al., 2010) for linear algebra, libcint (Sun, 2015) for integral evaluation, and libecpint (R. A. Shaw & Hill, 2017; R. Shaw & Hill, 2021) for effective core potentials.

Performance

Native builds provide performance comparable to established quantum chemistry packages through optimized integral evaluation, SIMD instructions, and parallel execution, while WebAssembly builds offer unprecedented accessibility with acceptable performance overhead for educational and prototyping use cases.

Availability and Documentation

OCC is available as open-source software under the GNU General Public License v3 at https://github.com/peterspackman/occ. Pre-built Python packages are available via PyPI

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(occpy). WebAssembly builds can be integrated via npm or used directly in browsers. Comprehensive continuous integration testing ensures reliability across Windows, macOS, and Linux platforms for all language bindings (C++, Python, JavaScript/WASM). Documentation at https://getocc.xyz includes interactive tutorials, where the code runs in the browser, complete C++ API references, and examples.

Conclusions

OCC provides a modern, accessible platform for quantum chemistry and crystallography that serves both as a research tool for method development and as a production-ready library powering applications like CrystalExplorer. Its unique combination of traditional electronic structure methods, crystallographic analysis, and WebAssembly support addresses key accessibility and integration challenges in computational chemistry. Future development will focus on expanding functional support, implementing excited state methods, and adding GPU acceleration.

Acknowledgements

OCC would not exist if it weren't for Tonto(Jayatilaka & Grimwood, 2003) its predecessor library and program at the backend of CrystalExplorer.

OCC builds upon a foundation of high-quality open-source libraries, and we gratefully acknowledge the developers and maintainers of these projects:

Core computational libraries: Eigen3 (Guennebaud et al., 2010) for linear algebra operations, libcint (Sun, 2015) for Gaussian integral evaluation, libxc (Lehtola et al., 2018) for exchange-correlation functionals, libecpint (R. Shaw & Hill, 2021) for effective core potential integrals, and gau2grid (Smith et al., 2025) for efficient grid evaluation of gaussians.

Crystallographic and optimization tools: gemmi (Wojdyr, 2022) for CIF file handling and crystallographic operations, and LBFGS++ (Qiu, 2025) for optimization algorithms. The molecular geometry optimization implementation was significantly informed by pyberny (Herrmann, 2019).

Infrastructure and utilities: fmt (Churaev, 2025) for string formatting, spdlog (Melman, 2025) for logging, oneTBB (*oneAPI Threading Building Blocks (oneTBB)*, 2025) for parallelization, CLI11 (Schreiner, 2025) for command-line parsing, nlohmann/json (Lohmann, 2025) for JSON handling, scnlib (Kosunen, 2025) for string parsing, and unordered_dense (Ankerl, 2025) for efficient hash containers.

Development tools: nanobind (Jakob, 2022) for Python bindings, Emscripten (Zakai, 2011) for WebAssembly compilation, and CPM.cmake (Melzer, 2024) for dependency management.

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